

# Technical Report

## Multi-Object Detection and Persistent ID Tracking in Soccer Footage

### 1. Objective

The objective was to develop a computer vision pipeline for detecting and tracking multiple subjects in public soccer footage while maintaining persistent IDs across video frames. The system addresses practical challenges such as occlusion, player overlap, similar appearance, motion, and camera movement.

### 2. Video Source

A publicly accessible soccer match video was selected for this assignment.

**Original Video Source:**

[Hugging Face — Bundesliga 2e57b9\\_0.mp4](#)

**Input Video:** soccer\_match.mp4

The selected footage contains multiple moving players and other soccer-related subjects, making it suitable for multi-object detection and tracking.

### 3. Detection Model

A football-specific YOLO model, football-player-detection.pt, was used for object detection.

The model detects the following classes:

| Class ID | Class      |
|----------|------------|
| 0        | Ball       |
| 1        | Goalkeeper |
| 2        | Player     |
| 3        | Referee    |

The detection threshold was set to 0.25 with an image size of 1280. YOLO was selected because the model is designed for football footage and provides bounding boxes suitable for tracking.



Figure 1: YOLO-based object detection on the selected soccer footage.

## 4. Tracking Algorithms

Two tracking approaches were implemented.

**ByteTrack** was used as the baseline tracker. It associates detections across consecutive frames and assigns tracking IDs to detected objects.

**BoT-SORT + Re-ID + GMC**: Used as the enhanced approach. BoT-SORT uses appearance-based Re-Identification to improve object association, while Global Motion Compensation using sparse optical flow helps handle camera movement.

## 5. Why This Combination Was Selected

YOLO performs detection but does not maintain identities between frames. A tracking algorithm is therefore required to associate detections over time.

ByteTrack provides a baseline for comparison, while BoT-SORT with Re-ID and GMC is better suited to challenging soccer scenes involving player overlap, similar appearances, temporary occlusion, and camera motion.

## 6. ID Consistency

Each detected subject receives a tracker-generated ID. Previous tracking information is used to associate new detections with existing tracks.

BoT-SORT additionally uses appearance-based Re-ID and GMC to improve associations during difficult tracking conditions. Tracking results were exported to CSV with frame number, track ID, class, confidence, and bounding-box coordinates.



Figure 2: BoT-SORT + Re-ID + GMC tracking with persistent IDs.

## 7. Results

| Metric                    | ByteTrack     | BoT-SORT + Re-ID + GMC |
|---------------------------|---------------|------------------------|
| Unique Tracks             | 73            | 63                     |
| Average Observations / ID | 240.62        | 278.00                 |
| Average Track Duration    | 243.60 frames | 280.38 frames          |
| Median Track Duration     | 31 frames     | 28 frames              |
| Longest Track             | 750 frames    | 750 frames             |

For this sequence, BoT-SORT produced fewer unique tracks and higher average observations and track duration, indicating higher measured track persistence in this experiment.

Ground-truth identity annotations were not available, so HOTA, IDF1, and ground-truth-based ID-switch metrics were not calculated.

## 8. Challenges and Failure Cases

The main challenges were:

- **Occlusion:** Players frequently overlap.
- **Similar appearance:** Players may have similar uniforms.
- **Camera motion:** Camera panning changes the apparent position of subjects.
- **Temporary detection loss:** Objects can disappear because of occlusion or detection uncertainty.
- **Track fragmentation:** Difficult associations can result in a new ID after an object is lost.

BoT-SORT's Re-ID and GMC components were used to improve robustness against these conditions.

## 9. Trajectory Visualization

As an optional enhancement, player trajectories were visualized using BoT-SORT tracking data. The bottom-center of each bounding box was used as the reference position, with a rolling history of recent positions.

The final video contains bounding boxes, tracking IDs, trajectories, active track count, and frame information.

Output: final\_submission\_video.mp4

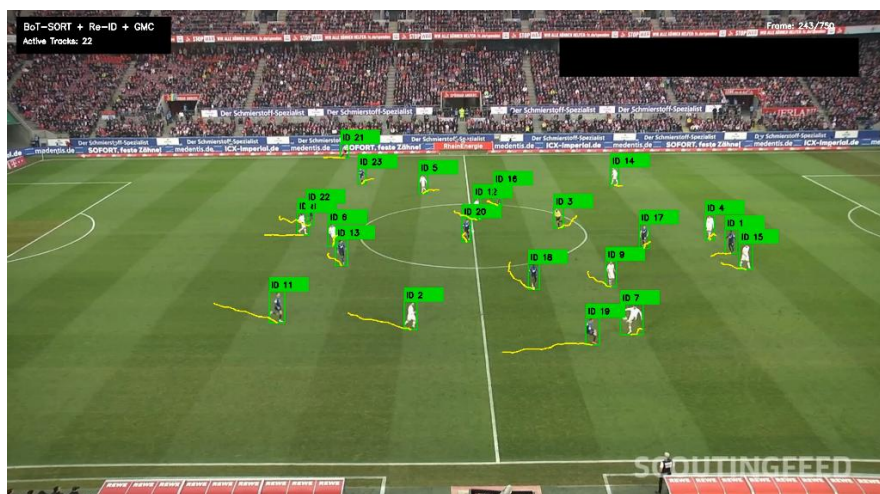


Figure 3: Final tracking output with player trajectories and persistent IDs.

## **10. Possible Improvements**

Future improvements include:

- Ground-truth-based evaluation using HOTA and IDF1
- Explicit identity-switch measurement
- Sports-specific Re-ID models
- Jersey-number recognition
- Team classification
- Player speed estimation
- Movement heatmaps
- Top-view/bird's-eye projection
- Real-time inference optimization

## **11. Conclusion**

The project implements an end-to-end soccer detection and persistent ID tracking pipeline using YOLO, ByteTrack, and BoT-SORT with Re-ID and GMC. For the selected sequence, BoT-SORT showed higher measured track persistence than ByteTrack. Trajectory visualization was also implemented as an optional enhancement.